

Midterm Exam #3: Z-Transform Solving

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Problem

Use the problem definitions given in the exam.

$$X(z) = \frac{z}{(z - \frac{1}{2})(z - 2)}$$

1.Poles and Zeros

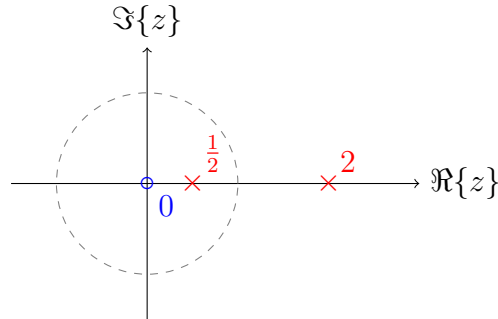
The zeros are the roots of the numerator $z = 0$. The poles are the roots of the denominator:

$$(z - \frac{1}{2})(z - 2) = 0 \implies z = \frac{1}{2}, 2.$$

Hence

$$\text{Zero: } z = 0, \quad \text{Poles: } z = \frac{1}{2}, 2.$$

Below is a TikZ diagram of the z -plane:



2. Region of Convergence (ROC)

The poles lie at $\frac{1}{2}$ and 2, so the ROC must be an annulus not containing any pole. There are three cases:

- $|z| > 2$: exterior of both poles (∞ included). This gives a *causal* (right-sided) $x[n]$.
- $|z| < \frac{1}{2}$: interior of both poles (0 included). This gives an *anti-causal* (left-sided) $x[n]$.
- $\frac{1}{2} < |z| < 2$: between the poles. This gives a *two-sided* (non-causal) $x[n]$.

3. Inverse Z-Transform

First do partial-fraction decomposition:

$$X(z) = \frac{z}{(z - \frac{1}{2})(z - 2)} = \frac{A}{z - \frac{1}{2}} + \frac{B}{z - 2}.$$

Solve

$$z = A(z - 2) + B(z - \frac{1}{2}) \implies A = -\frac{1}{3}, B = \frac{4}{3}.$$

Thus

$$X(z) = -\frac{1}{3} \frac{1}{z - \frac{1}{2}} + \frac{4}{3} \frac{1}{z - 2}.$$

3.1 Causal ROC: $|z| > 2$

Use $\frac{1}{z-a} = \sum_{n=0}^{\infty} a^n z^{-n-1}$, $|z| > |a|$.

$$\frac{1}{z - \frac{1}{2}} = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n-1}, \quad \frac{1}{z - 2} = \sum_{n=0}^{\infty} 2^n z^{-n-1}.$$

Hence

$$X(z) = -\frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n-1} + \frac{4}{3} \sum_{n=0}^{\infty} 2^n z^{-n-1}.$$

Shift index $m = n + 1$:

$$X(z) = \sum_{m=1}^{\infty} \left(-\frac{1}{3} \left(\frac{1}{2}\right)^{m-1} + \frac{4}{3} 2^{m-1} \right) z^{-m}.$$

Thus

$$x[n] = \begin{cases} -\frac{1}{3} \left(\frac{1}{2}\right)^{n-1} + \frac{4}{3} 2^{n-1}, & n \geq 1, \\ 0, & n \leq 0. \end{cases}$$

3.2 Anti-Causal ROC: $|z| < \frac{1}{2}$

Use $\frac{1}{z-a} = -\frac{1}{a} \sum_{n=0}^{\infty} (z/a)^n$, $|z| < |a|$.

$$\frac{1}{z - \frac{1}{2}} = -2 \sum_{n=0}^{\infty} (2z)^n, \quad \frac{1}{z - 2} = -\frac{1}{2} \sum_{n=0}^{\infty} (z/2)^n.$$

Multiply by coefficients:

$$-\frac{1}{3} \frac{1}{z - \frac{1}{2}} = \frac{2}{3} \sum_{n=0}^{\infty} 2^n z^n, \quad \frac{4}{3} \frac{1}{z - 2} = -\frac{2}{3} \sum_{n=0}^{\infty} 2^{-n} z^n.$$

So

$$X(z) = \sum_{n=0}^{\infty} \left(\frac{2}{3} 2^n - \frac{2}{3} 2^{-n} \right) z^n = \sum_{n=0}^{\infty} a_n z^n,$$

with $a_n = \frac{2}{3}(2^n - 2^{-n})$. Since $z^n = z^{-(-n)}$, we read off

$$x[k] = a_{-k} = \frac{2}{3}(2^{-k} - 2^k), \quad k \leq 0,$$

and $x[n] = 0$ for $n > 0$. Equivalently,

$$x[n] = \begin{cases} \frac{2}{3}(2^{-n} - 2^n), & n \leq 0, \\ 0, & n > 0. \end{cases}$$

4. Extra: BIBO Stability

I wanted to add this part, just I believe it is important to check this for a system. Here our causal part is the $|z| > 2$ since our absolute bigger pole is 2. and below $|z| < \frac{1}{2}$ is the anti-causal part for ROC, in between is non-causal, so both past and future values are used for our current input. For the BIBO stability we need to have every pole inside unit circle in complex domain, for causal systems. Which we do not. For anti-causal, all should be out of the unit circle. For non-causal the area between poles should be including unit circle, hence only our non-causal system is BIBO stable. Important since, when one is implementing a system, they should know their properties and what they will expect.